Method report for Austrian Values in Crisis (VIC) data for Wave 1 (2020), Wave 2 (2021), Wave 3 (2022) and the panel data

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1. Introduction

The following document describes the sampling design and the weighting strategy for the three waves of the Austrian Values in Crisis (VIC Austria) survey. The sample includes all respondents from wave 1 (W1, n=2,018), from wave 2 (W2, n=2,082) and wave 3 (W3, n=2,123). As the panel data includes only a subset of the entire dataset, four panel samples are differentiated, namely panel data W12 (n=1,222) including respondents, who participated in W1 and W2, panel data W13 (n=832) covering all respondents, who participated in W1 and W3, panel data W23 (n=1,119) defining a subsample with all participants of W2 and W3 and finally the panel data W123 (n=747) consisting of participants of all three waves. The dataset contains filter variables for each subset.

Additionally, commonly used scales are prepared such as the occupational classification with the ISCO coding, the Inglehard-Index (all three waves), the Schwartz Values (all three waves), one personality scale (wave 1) and the POLID Ideology Scores (wave 2 and wave 3).

The report consists of five different parts. The section on sampling (part 2) describes the main features of the online-access panel study. The third section presents all relevant information concerning the weighting including the comparison of the raw data of the various subsamples of the dataset and their respective target populations. Furthermore, details on the applied sampling strategy and the computation of weights corresponding to the respective samples (W1, W2, W3, W12, W13, W23, W123) are provided. Section four provides information on how to use the ISCO variables included in the dataset. Finally, the fifth section introduces commonly used scales operationalising latent constructs such as values, personality or ideology scores, which were prepared for the dataset users.

Besides those additional scales, the VIC study integrates many item batteries referring to attitudes towards politics and perceptions on public health, as well as the own exposure to the pandemic and perceptions of social and economic consequences for the Austrian society. The three waves have many repeating modules, but some specific questions are only partially included across the waves.[2]

[2] Please see questionnaires which are also part of this release to get an overview of the survey.

2. What population does VIC Austria represent?

The survey was conducted by the Marketagent.com online reSEARCH GmbH (Marketagent)[3 4 5] based on a consumer online panel with 139,500 participants in Austria. The online panel integrates persons who are at least 14 years or older. All educational groups, professions and Austrian regions are well represented however people in older age groups and immigrants are underrepresented. The participants speak the official language of the country (regardless of nationality and citizenship) and do not live either abroad or in institutions such as prisons and hospitals during the entire fieldwork period.

Table 1: Comparison sampling frame and target population for 2020, 2021 and 2022

	Characteristic	Market-agent Panel 2020[1]	Market-agent Panel 2021[1]	Market-agent Panel 2022[1]	Statistics Austria 2020[2]	Statistics Austria 2021[2]	Statistics Austria 2022[2]
Female	14-19 years	2.8	1.5	1.8	3.3	3.3	3.3
	20-29 years	15.7	14.8	14.3	7.1	6.9	6.8
	30-39 years	16.8	17.6	17.3	7.8	7.8	7.8
	40-49 years	9.9	10.7	11.3	7.7	7.6	7.6
	50-59 years	7.1	7.4	7.5	9.1	9.0	9.0
	60+ years	3.8	4.5	5.0	16.2	16.4	16.6
	Sum	56.2	56.6	57.3	51.2	51.1	51.0
Male	14-19 years	2.6	1.3	1.3	3.5	3.5	3.5
	20-29 years	12.0	11.4	10.9	7.5	7.3	7.2
	30-39 years	10.6	11.4	11.0	8.0	8.0	8.1
	40-49 years	7.6	7.8	7.8	7.7	7.6	7.6
	50-59 years	6.0	6.1	6.0	9.1	9.0	8.9
	60+ years	5.1	5.4	5.7	13.1	13.4	13.6
	Sum	43.8	43.4	42.7	48.8	48.9	49.0
Marital status	Single	58.3	54.4	58.3	30.7	30.7	n/a
	Married/partnership	30.6	35.5	36.4	45.8	45.7	n/a
	Divorced	9.9	9.0	8.9	13.7	13.9	n/a
	Widowed	1.2	1.1	1.1	9.8	9.7	n/a
Household size	Single person	20.3	20.4	20.3	37.8	38.0	n/a
	Two persons	33.2	33.3	33.6	30.4	30.4	n/a
	Three and more	46.5	46.3	46.5	31.9	31.6	n/a
Children per household	None	56.5	57.0	56.7	77.4	77.4	n/a
	One	35.2	34.9	35.2	10.8	11.0	n/a
	Two or more	8.3	8.1	8.1	11.8	11.7	n/a

Notes: [1] retrieved from Marketagent Panel Books 2020[5], 2021[4] and 2022[5]; [2] Statistics Austria relies on different databases: gender, age (Statistics Austria 2022)[6]; Marital status, Household size and Children in Household (Statistics Austria 2021)[7]

[3] Marketagent.com. (2020). *Panel Book* [Brochure]. Baden, Austria. https://b2b.marketagent.com/media/itkf4i0o/marketagent-panel-book-2020.pdf (last accessed 2020)

[4] Marketagent.com. (2021). *Panel Book* [Brochure]. Baden, Austria. https://b2b.marketagent.com/media/itkf4i0o/marketagent-panel-book-2021.pdf (last accessed 07/26/2021)

[5] Marketagent.com. (2022). *Panel Book* [Brochure]. Baden, Austria: Author. https://b2b.marketagent.com/media/5jrj43nr/marketagent-panel-book-2022.pdf (last accessed 01/04/2023)

[6] Statistics Austria (2022). STATcube – statistical database of Statistics Austria. Population at the beginning of the year from 2002 onwards (uniform territorial status 2020; data are continuously imported into a population statistics database system), based on the Central Register of Residents. https://statcube.at/statistik.at/ext/statcube/jsf/terms.xhtml (last accessed 10/13/2022)

[7] Statistics Austria (2021). STATcube – statistical database of Statistics Austria. Private households Microcensus

Table 1 compares the samples of the online panel with official statistics by Statistics Austria. Compared to the target population, the Marketagent sample consists of disproportionally more women and people between 20 and 39 years. Additionally, samples drawn from the panel recruiting people 60 years and more will rely on a smaller pool than younger age groups as the Marketagent online access panel includes fewer older people. Similar deviations from the target population appear regarding marital status, household size and number of children in the household. Married and widowed people are underrepresented as well as people in single person households and households without children. Obviously, these deviations relate to the age distortion as older people are more likely to be widowed and to live in a single person household without children.

3. The weighting procedure in the Austrian VIC Survey

The VIC Austria is part of a cross-national survey and as most modern population surveys, complex sampling strategies are applied to improve survey quality. Considering these sampling procedures is particularly important for cross-national surveys like the VIC study in order to mirror country characteristics adequately.[8] A commonly applied method is the combination of quota sampling and poststratification weighting based on register data to correct for the different sampling probabilities.[9] [10]

In the online panel a stratified quota sampling strategy was applied for gender, age, region and education. The distribution of the raw data of W1, W2 and W3 is presented in Table 2 and the panel data for W12, W13, W23 and W123 in Table 3.

As a result of the quotas, the sample represents the target population rather well already. Nevertheless, the sampling probabilities for the different groups vary due to the mismatch between sampling frame and target population. This will especially impose a bias for the age group 70+ and lower educational groups. A certain coverage bias will be present in the sample as the selection into the online access panel is not random.

To correct for deviations between sample and target population, survey sample weights were calculated by raking aligning the data to population margins. The reference population margins are gender, age and education and region separately (see Table 2 and Table 3).

Labor force survey since 2020 (uniform territorial status 2021; data are continuously imported into a population statistics database system), based on the Central Register of Residents. https://statcube.at/statistik.at/ext/statcube/jsf/tableView/tableView.xhtml (last accessed 01/13/2023)

[8] Kish, L. (1994). Multipopulation Survey Designs: Five Types with Seven Shared Aspects. International Statistical Review / Revue Internationale de Statistique, 62(2), 167–186. JSTOR. https://doi.org/10.2307/1403507

[9] DeBell, M., & Krosnick, J. A. (2009). Computing Weights for American National Election Study Survey Data. https://electionstudies.org/wp-content/uploads/2018/04/nes012427.pdf (last accessed 07/27/2021)

[10] Lynn, P., & Anghelescu, G. (2018). Europoean Social Survey Round 8 Weighting Strategy. https://www.europeansocialsurvey.org/sites/default/files/2023-06/ESS8_weighting_strategy_0.pdf (last accessed 12/05/2023)

Table 2: Comparison unweighted (raw) sample Wave 1 to 3 and target population.

	Characteristic	Raw Wave 1	Statistics Austria 2020	Raw Wave 2	Statistics Austria 2021	Raw Wave 3	Statistics Austria 2022
Female[1]	14-19 years	3.1	3.3	3.1	3.3	3.3	3.3
	20-29 years	8.1	7.1	8.2	6.9	8.3	6.8
	30-39 years	7.9	7.8	8.0	7.8	8.1	7.8
	40-49 years	8.7	7.7	8.4	7.6	8.6	7.6
	50-59 years	9.4	9.1	9.3	9.0	9.6	9.0
	60-69 years	7.2	6.8	7.3	7.0	7.4	7.2
	70+ years	6.3	9.4	6.0	9.4	5.8	9.4
	Sum	50.8	51.2	50.5	51.1	50.9	51.0
Male[1]	14-19 years	2.9	3.5	2.6	3.5	2.0	3.5
	20-29 years	8.5	7.5	8.4	7.3	8.5	7.2
	30-39 years	7.8	8.0	7.7	8.0	8.1	8.1
	40-49 years	8.4	7.7	8.4	7.6	8.5	7.6
	50-59 years	9.9	9.1	9.6	9.0	9.5	8.9
	60-69 years	6.2	6.3	7.8	6.5	6.9	6.8
	70+ years	5.5	6.8	5.1	6.8	5.6	6.9
	Sum	49.2	48.8	49.5	48.9	49.1	49.0
Region[1]	Vorarlberg	4.4	4.5	4.0	4.5	3.9	4.4
	Tyrol	8.6	8.5	8.1	8.5	7.7	8.5
	Salzburg	6.2	6.3	6.3	6.3	5.0	6.2
	Styria	14.3	14.0	14.3	14.0	14.5	14.1
	Carinthia	6.5	6.3	6.3	6.3	5.9	6.4
	Upper Austria	16.9	16.7	17.5	16.7	15.8	16.6
	Lower Austria	18.9	18.9	18.9	18.9	21.3	18.9
	Vienna	20.5	21.5	20.9	21.5	22.4	21.5
	Burgenland	3.7	3.3	3.6	3.3	3.4	3.4
Education[2]	Compulsory education	17.2	24.8	14.0	24.8	13.7	24.8
	Apprenticeship	38.7	31.1	41.8	31.1	38.1	31.1
	Vocational middle school	13.1	13.8	14.8	13.8	13.9	13.8
	General or vocational secondary school	19.7	15.5	18.7	15.5	20.3	15.5
	University	11.4	14.9	10.8	14.9	14.0	14.9

Notes: [1] retrieved from Statistics Austria 2022[6]; [2] retrieved from Statistics Austria 2020[11]

[11] Statistics Austria (2020). Educational attainment of the population in 2020 aged 15 and over. https://www.statistik.at/fileadmin/pages/315/3_Bildungsstand_ab15_2020.ods (last accessed 10/13/2022)

Table 3: Comparison unweighted (raw) sample and target population for panel data

	Characteristic	Raw Panel W12	Statistics Austria 2021	Raw Panel W13	Raw Panel W23	Raw Panel W123	Statistics Austria 2022
Female[1]	14-19 years	1.7	3.3	1.1	1.4	0.5	3.3
	20-29 years	4.5	6.9	4.1	5.7	3.2	6.8
	30-39 years	7.5	7.8	7.0	7.5	6.7	7.8
	40-49 years	9.1	7.6	9.0	8.8	9.0	7.6
	50-59 years	10.7	9.0	11.9	11.6	12.5	9.0
	60-69 years	9.1	7.0	10.4	10.0	11.4	7.2
	70+ years	7.5	9.4	8.2	7.3	8.2	9.4
	Sum	50.2	51.1	51.7	52.2	51.5	51.0
Male[1]	14-19 years	1.1	3.5	0.5	0.5	0.4	3.5
	20-29 years	5.9	7.3	5.3	4.6	4.2	7.2
	30-39 years	6.9	8.0	7.1	7.8	7.1	8.1
	40-49 years	8.7	7.6	8.8	9.1	8.9	7.6
	50-59 years	12.2	9.0	12.5	11.3	12.9	8.9
	60-69 years	8.4	6.5	8.2	8.8	8.7	6.8
	70+ years	6.6	6.8	5.9	5.7	6.3	6.9
	Sum	49.8	48.9	48.3	47.8	48.5	49.0
Region[1]	Vorarlberg	4.3	4.5	4.7	4.3	4.8	4.4
	Tyrol	8.3	8.5	9.0	8.6	9.1	8.5
	Salzburg	6.0	6.3	5.3	5.9	5.2	6.2
	Styria	15.2	14.0	14.8	14.5	15.0	14.1
	Carinthia	5.9	6.3	5.2	6.0	5.1	6.4
	Upper Austria	16.4	16.7	16.8	17.6	16.3	16.6
	Lower Austria	19.8	18.9	19.7	18.0	19.1	18.9
	Vienna	20.5	21.5	20.8	22.0	21.8	21.5
	Burgenland	3.5	3.3	3.7	3.2	3.5	3.4
Education[2]	Compulsory education	16.0	24.8	15.9	12.6	15.4	24.8
	Apprenticeship	40.4	31.1	40.6	43.7	39.9	31.1
	Vocational middle school	14.1	13.8	15.1	16.6	15.9	13.8
	General or vocational secondary school	18.0	15.5	16.9	16.7	17.0	15.5
	University	11.5	14.9	11.4	10.4	11.8	14.9

Notes: [1] retrieved from Statistics Austria 2022[8]; [2] retrieved from Statistics Austria 2020[13]

The R package anesrake by Josh Pasek (2018) was used to estimate the weights.[12] The calculated weight variables for the different samples have a minimum ranging from 0.458 (W123 to 0.593 (W1) and a maximum ranging from 2.262 (W1) to 7.619 (W123). In no sample are the weights below 0.3. For W1 and W12 no weight exceeded 3 and for the other subsample a maximum of 12 cases (W13) exceeded 3. The respective sum of the weights is reported in Table 4. Weighting scores are considered moderately skewed, and no adjustment of the weights was made to account for outliers.[13] Figure 1 presents the distributions of the weights. The single wave subsamples (W1, W2, W3) and the panel sample W12 perform best with the smallest weights only partially exceeding weights of 3. As the single wave subsamples (W1, W2, W3) have been sampled according to quotas with respect to gender, age, education and region, they already represent the population well. The sampling of Wave 2 and

[12] Pasek, J. (2018). anesrake R package version 0.8. https://cran.r-project.org/web/packages/anesrake/anesrake.pdf (last accessed 01/21/2023)

[13] Little, R. J. A., Lewitzky, S., Heeringa, S., Lepkowski, J., & Kessler, R. C. (1997). Assessment of Weighting Methodology for the National Comorbidity Survey. American Journal of Epidemiology, 146(5), 439–449. https://doi.org/10.1093/oxfordjournals.aje.a009297

Wave 3 included the respondents of previous waves and selection is partially based on the willingness of respondents from Wave 1 or Wave 2 to participate in Wave 2 or Wave 3 respectively. Hence, the W2 and W3 samples deviate stronger from the population leading to larger weights. The panel data for W13, W23 and W123 consists of fewer cases due to panel mortality. Nonetheless, the weights are comparably small considering the reduced sample size.

Figure 1: Distribution of the weights

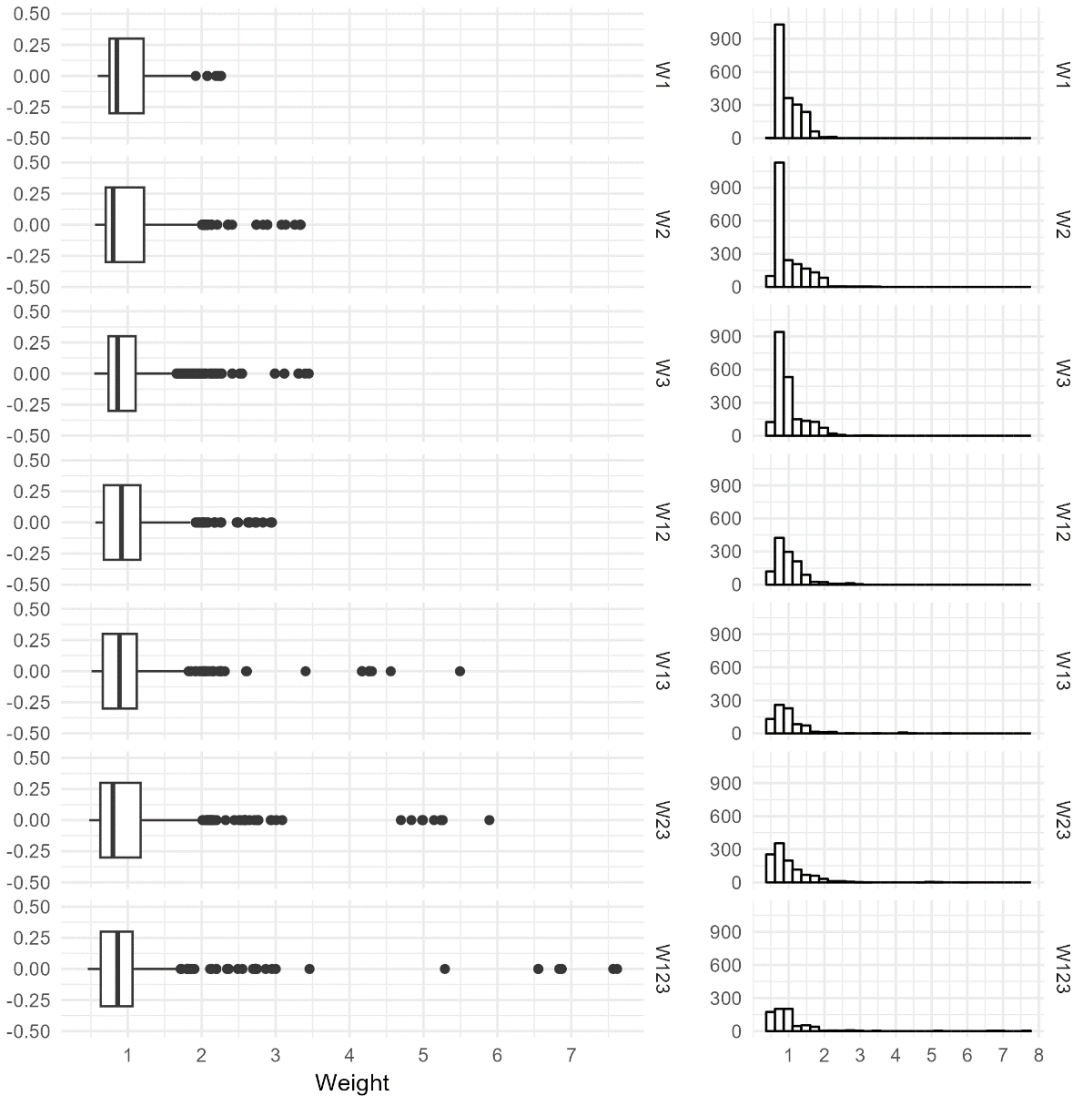


Table 4: Descriptives of weighting variables

Parameter	W1	W2	W3	W12	W13	W23	W123
n	2,018	2,082	2,123	1,222	832	1,119	747
Standard deviation	0.308	0.424	0.408	0.411	0.543	0.592	0.711
1st Quartil	0.753	0.702	0.739	0.675	0.662	0.629	0.635
Median	0.854	0.803	0.861	0.914	0.889	0.799	0.862
3rd Quartil	1.213	1.222	1.107	1.172	1.123	1.175	1.064
Minimum	0.593	0.557	0.548	0.564	0.509	0.481	0.458
Maximum	2.262	3.338	3.449	2.949	5.494	5.891	7.619
Design effect	1.095	1.180	1.166	1.168	1.294	1.350	1.505
No. of weights > 3 (W>3)	0	6	7	0	12	11	9
Sum of W>3	0.000	19.476	23.020	0.000	51.597	52.122	53.748
Design effect	1.095	1.180	1.166	1.168	1.294	1.350	1.505

Table 5 compares the raw data and the adjusted (weighted) data of all three waves of the VIC survey. It gets visible that the adjusted weighted data are consistent with the official statistics. This means that the VIC data can be considered as representative for the population (at least when it comes to the weighted distribution of gender, age, education and region).

Table 5: Comparison of unweighted (raw) and weighted (adjusted) data for Wave 1 to 3

	Characteristic	**Wave 1**		**Wave 2**		**Wave 3**	
		raw	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**
Gender[1]	Female	50.8	51.2	50.5	51.1	50.9	51.0
	Male	49.2	48.8	49.5	48.9	49.1	49.0
Age[1]	14-19 years	6.0	6.8	5.8	6.7	5.3	6.7
	20-29 years	16.6	14.5	16.7	14.3	16.7	14.0
	30-39 years	15.8	15.8	15.8	15.9	16.2	15.9
	40-49 years	17.1	15.4	16.8	15.3	17.0	15.2
	50-59 years	19.3	18.1	18.9	18.1	19.1	17.9
	60-69 years	13.4	13.2	15.0	13.6	14.2	14.0
	70+ years	11.8	16.1	11.1	16.2	11.4	16.3
Region[1]	Vorarlberg	4.4	4.5	4.0	4.5	3.9	4.4
	Tyrol	8.6	8.5	8.1	8.5	7.7	8.5
	Salzburg	6.2	6.3	6.3	6.3	5.0	6.2
	Styria	14.3	14.0	14.3	14.0	14.5	14.1
	Carinthia	6.5	6.3	6.3	6.3	5.9	6.4
	Upper Austria	16.9	16.7	17.5	16.7	15.8	16.6
	Lower Austria	18.9	18.9	18.9	18.9	21.3	18.9
	Vienna	20.5	21.5	20.9	21.5	22.4	21.5
	Burgenland	3.7	3.3	3.6	3.3	3.4	3.4
Education[2]	Compulsory education	17.2	24.8	14.0	24.8	13.7	24.8
	Apprenticeship	38.7	31.1	41.8	31.1	38.1	31.1
	Vocational middle school	13.1	13.8	14.8	13.8	13.9	13.8
	General or vocational secondary school	19.7	15.5	18.7	15.5	20.3	15.5
	University	11.4	14.9	10.8	14.9	14.0	14.9

The same is true when it comes to the panel data of people participating in W12, W13, W23 and W123. Also, these longitudinal data should be weighted to derive generalized conclusions for societal changes in Austria over these specific time frames.

Table 6: Comparison of unweighted (raw) and weighted (adjusted) data for Wave 1 to 3

	Characteristics	Panel W12		Panel W13		Panel W23		Panel W123	
		raw	adj.	raw	adj.	raw	adj.	raw	adj.
Gender	Female	50.2	51.2	51.7	51.2	52.2	51.1	51.5	51.2
	Male	49.8	48.8	48.3	48.8	47.8	48.9	48.5	48.8
Age	14-19 years	2.9	6.8	1.7	6.8	2.1	6.7	1.1	6.8
	20-29 years	10.4	14.5	9.4	14.5	10.4	14.3	7.4	14.5
	30-39 years	14.3	15.8	14.1	15.8	15.4	15.9	13.8	15.8
	40-49 years	17.8	15.4	17.8	15.4	17.8	15.3	17.8	15.4
	50-59 years	22.9	18.1	24.4	18.1	22.8	18.1	25.3	18.1
	60-69 years	17.4	13.2	18.5	13.2	18.7	13.6	20.1	13.2
	70+ years	14.2	16.1	14.2	16.1	13.0	16.2	14.6	16.1
Region	Vorarlberg	4.3	4.5	4.7	4.5	4.3	4.5	4.8	4.5
	Tyrol	8.3	8.5	9.0	8.5	8.6	8.5	9.1	8.5
	Salzburg	6.0	6.3	5.3	6.3	5.9	6.3	5.2	6.3
	Styria	15.2	14.0	14.8	14.0	14.5	14.0	15.0	14.0
	Carinthia	5.9	6.3	5.2	6.3	6.0	6.3	5.1	6.3
	Upper Austria	16.4	16.7	16.8	16.7	17.6	16.7	16.3	16.7
	Lower Austria	19.8	18.9	19.7	18.9	18.0	18.9	19.1	18.9
	Vienna	20.5	21.5	20.8	21.5	22.0	21.5	21.8	21.5
	Burgenland	3.5	3.3	3.7	3.3	3.2	3.3	3.5	3.3
Education	Compulsory education	16.0	24.8	15.9	24.8	12.6	24.8	15.4	24.8
	Apprenticeship	40.4	31.1	40.6	31.1	43.7	31.1	39.9	31.1
	Vocational middle school	14.1	13.8	15.1	13.8	16.6	13.8	15.9	13.8
	General or vocational secondary school	18.0	15.5	16.9	15.5	16.7	15.5	17.0	15.5
	University	11.5	14.9	11.4	14.9	10.4	14.9	11.8	14.9

Table 7 to Table 13 compare the weighted data in a three-way table of gender, age and education with the unweighted data. The largest deviations between weighted and unweighted distribution can be observed in the columns “compulsory school” and “apprenticeship”. The deviations in the other groups are minor. In general, the adjustments made by the weights are considerably small and should be unproblematic with respect to potential adjustment errors.

Table 7: Three-way table with shares for gender, age and education comparing raw (unweighted) and adjusted (weighted) data for Wave 1

		Highest completed training (education)											
		Compulsory school		**Apprenticeship**		**Vocational middle school**		**General or vocational secondary school**		**University**		***Total***	
Gender	**Age Groups**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	***raw***	***adjusted***
Female	**< 20 years**	2.2	3.0	0.2	0.2	0.0	0.0	0.5	0.4	0.0	0.0	*3.1*	*3.6*
	20-29 years	0.8	1.2	3.7	2.7	0.6	0.6	2.2	1.6	0.7	0.8	*8.1*	*6.9*
	30-39 years	1.5	2.3	3.9	3.1	0.9	1.0	0.9	0.7	0.7	0.8	*7.9*	*7.9*
	40-49 years	2.0	2.7	3.0	2.2	1.3	1.2	1.1	0.8	1.2	1.3	*8.7*	*8.1*
	50-59 years	1.4	2.0	2.7	2.1	2.0	2.0	1.9	1.3	1.4	1.6	*9.4*	*9.0*
	60-69 years	0.9	1.5	3.3	2.7	2.2	2.3	0.4	0.3	0.3	0.4	*7.2*	*7.2*
	70+ years	0.5	1.1	1.6	1.8	1.2	1.8	1.9	2.1	1.0	1.8	*6.3*	*8.6*
Male	**< 20 years**	1.8	2.4	0.3	0.2	0.1	0.1	0.7	0.5	0.0	0.0	*2.9*	*3.2*
	20-29 years	1.5	2.2	2.6	1.9	0.6	0.6	3.2	2.3	0.6	0.8	*8.5*	*7.7*
	30-39 years	1.5	2.3	3.2	2.5	0.6	0.7	1.3	1.0	1.1	1.4	*7.8*	*8.0*
	40-49 years	1.2	1.7	4.2	3.0	0.7	0.7	1.0	0.7	1.1	1.2	*8.3*	*7.3*
	50-59 years	1.0	1.5	5.0	3.9	0.9	1.0	1.8	1.4	1.2	1.5	*9.9*	*9.2*
	60-69 years	0.4	0.7	3.3	2.8	1.1	1.2	0.9	0.8	0.4	0.6	*6.2*	*6.0*
	70+ years	0.1	0.3	1.6	1.9	0.5	0.7	1.6	1.8	1.5	2.7	*5.5*	*7.5*
Total		*17.1*	*24.8*	*38.7*	*31.1*	*13.1*	*13.8*	*19.7*	*15.5*	*11.4*	*14.8*	*100.0*	*100.0*

Table 8: Three-way table with shares for gender, age and education comparing raw (unweighted) and adjusted (weighted) data for Wave 2

		Highest completed training (education)											
		Compulsory school		**Apprenticeship**		**Vocational middle school**		**General or vocational secondary school**		**University**		***Total***	
Gender	**Age Groups**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	***raw***	***adjusted***
Female	**< 20 years**	1.5	2.4	0.0	0.0	0.1	0.1	1.3	1.0	0.1	0.2	*3.1*	*3.6*
	20-29 years	0.7	1.1	2.6	1.7	0.6	0.4	2.8	1.9	1.5	1.7	*8.2*	*6.8*
	30-39 years	0.9	1.8	4.5	3.4	0.9	0.9	1.1	1.0	0.6	0.9	*8.0*	*8.0*
	40-49 years	1.7	2.8	3.7	2.4	1.2	0.9	0.8	0.6	1.0	1.1	*8.4*	*7.8*
	50-59 years	1.4	2.6	3.4	2.5	2.1	1.8	1.3	1.0	1.2	1.5	*9.3*	*9.3*
	60-69 years	1.1	2.0	3.7	2.6	1.9	1.6	0.4	0.3	0.2	0.3	*7.3*	*6.8*
	70+ years	0.4	1.2	2.0	2.3	1.3	1.8	1.4	1.8	0.9	1.8	*6.0*	*8.8*
Male	**< 20 years**	1.3	2.1	0.4	0.2	0.0	0.0	0.8	0.6	0.1	0.1	*2.6*	*3.0*
	20-29 years	1.3	2.1	2.4	1.5	0.9	0.6	2.9	2.1	0.9	1.1	*8.4*	*7.4*
	30-39 years	0.9	1.8	4.1	3.2	0.8	0.7	0.9	0.8	1.0	1.5	*7.7*	*7.9*
	40-49 years	1.2	2.0	3.9	2.6	1.1	0.9	1.1	0.8	1.1	1.3	*8.4*	*7.5*
	50-59 years	0.9	1.6	5.2	3.8	1.3	1.2	1.5	1.2	0.8	1.0	*9.6*	*8.8*
	60-69 years	0.5	1.0	4.0	2.9	1.6	1.4	1.3	1.0	0.4	0.5	*7.8*	*6.8*
	70+ years	0.1	0.4	1.8	2.1	1.1	1.6	1.1	1.4	0.9	1.8	*5.1*	*7.4*
Total		*14.0*	*24.8*	*41.8*	*31.1*	*14.8*	*13.9*	*18.7*	*15.5*	*10.7*	*14.8*	*100.0*	*100.0*

Table 9: Three-way table with shares for gender, age and education comparing raw (unweighted) and adjusted (weighted) data for Wave 3

		Highest completed training (education)											
		Compulsory school		**Apprenticeship**		**Vocational middle school**		**General or vocational secondary school**		**University**		***Total***	
Gender	**Age Groups**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	***raw***	***adjusted***
Female	**< 20 years**	1.5	2.7	0.5	0.4	0.1	0.1	1.1	0.8	0.1	0.1	*3.3*	*4.1*
	20-29 years	0.6	0.9	2.4	1.7	0.6	0.5	3.0	1.9	1.7	1.5	*8.3*	*6.5*
	30-39 years	1.1	2.0	3.0	2.3	0.9	0.8	1.2	0.8	1.8	1.8	*8.1*	*7.8*
	40-49 years	1.6	2.5	3.4	2.4	0.9	0.8	1.1	0.7	1.6	1.4	*8.6*	*7.9*
	50-59 years	1.5	2.5	3.6	2.8	2.2	1.9	1.3	0.9	1.0	1.0	*9.6*	*9.1*
	60-69 years	1.1	2.1	3.0	2.4	1.5	1.4	1.2	0.9	0.6	0.6	*7.4*	*7.4*
	70+ years	0.6	1.6	1.6	1.9	1.3	1.9	1.5	1.6	0.9	1.3	*5.8*	*8.3*
Male	**< 20 years**	0.9	1.6	0.3	0.3	0.0	0.1	0.8	0.6	0.0	0.0	*2.0*	*2.6*
	20-29 years	1.3	2.1	2.6	1.9	0.5	0.4	2.6	1.8	1.4	1.3	*8.5*	*7.5*
	30-39 years	1.0	1.9	3.5	2.9	1.1	1.0	1.3	1.0	1.1	1.2	*8.1*	*8.1*
	40-49 years	0.7	1.1	3.5	2.6	1.2	1.0	1.5	1.0	1.6	1.6	*8.5*	*7.4*
	50-59 years	1.1	2.0	4.6	3.6	1.5	1.4	1.4	1.0	0.8	0.8	*9.5*	*8.8*
	60-69 years	0.4	0.7	3.8	3.3	0.9	0.9	1.0	0.8	0.8	0.9	*6.9*	*6.6*
	70+ years	0.3	0.9	2.1	2.6	1.1	1.7	1.4	1.6	0.7	1.2	*5.6*	*8.0*
Total		*13.7*	*24.8*	*38.2*	*31.1*	*13.9*	*13.9*	*20.3*	*15.4*	*13.9*	*14.8*	*100.0*	*100.0*

Table 10: Three-way table with shares for gender, age and education comparing raw (unweighted) and adjusted (weighted) data for panel data Wave 1 and Wave 2 (W12)

		Highest completed training (education)											
		Compulsory school		**Apprenticeship**		**Vocational middle school**		**General or vocational secondary school**		**University**		***Total***	
Gender	**Age Groups**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	***raw***	***adjusted***
Female	**< 20 years**	1.2	3.3	0.2	0.4	0.0	0.0	0.2	0.4	0.0	0.0	*1.7*	*4.0*
	20-29 years	0.4	0.9	2.5	2.9	0.3	0.5	0.8	0.9	0.5	1.0	*4.5*	*6.2*
	30-39 years	1.6	2.5	3.4	2.8	0.9	1.1	0.9	0.8	0.7	1.0	*7.5*	*8.2*
	40-49 years	2.2	2.7	3.3	2.3	1.4	1.3	1.1	0.7	1.1	1.3	*9.1*	*8.2*
	50-59 years	1.9	2.1	3.1	2.0	2.3	2.0	2.1	1.3	1.4	1.4	*10.7*	*8.8*
	60-69 years	1.1	1.3	4.3	2.6	2.9	2.4	0.5	0.3	0.3	0.3	*9.1*	*7.1*
	70+ years	0.5	0.9	2.1	1.9	1.4	1.7	2.3	2.2	1.3	1.9	*7.5*	*8.6*
Male	**< 20 years**	0.7	1.8	0.2	0.3	0.1	0.2	0.2	0.4	0.0	0.0	*1.1*	*2.6*
	20-29 years	1.3	2.7	1.7	2.0	0.1	0.1	2.5	3.1	0.2	0.5	*5.9*	*8.4*
	30-39 years	1.5	2.3	3.1	2.7	0.5	0.6	0.7	0.6	1.1	1.6	*6.9*	*7.7*
	40-49 years	1.3	1.6	4.1	2.7	0.7	0.7	1.2	0.8	1.3	1.4	*8.7*	*7.2*
	50-59 years	1.2	1.4	5.9	3.8	1.1	1.0	2.5	1.6	1.4	1.5	*12.2*	*9.4*
	60-69 years	0.7	0.8	4.3	2.7	1.7	1.5	1.1	0.7	0.5	0.5	*8.4*	*6.2*
	70+ years	0.2	0.4	2.3	2.1	0.7	0.8	1.8	1.7	1.6	2.4	*6.6*	*7.4*
Total		*15.9*	*24.7*	*40.4*	*31.1*	*14.1*	*13.9*	*18.0*	*15.5*	*11.5*	*14.8*	*100.0*	*100.0*

Table 11: Three-way table with shares for gender, age and education comparing raw (unweighted) and adjusted (weighted) data for panel data Wave 1 and Wave 3 (W13)

		Highest completed training (education)											
		Compulsory school		**Apprenticeship**		**Vocational middle school**		**General or vocational secondary school**		**University**		***Total***	
Gender	**Age Groups**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	***raw***	***adjusted***
Female	**< 20 years**	1.0	4.1	0.1	0.3	0.0	0.0	0.0	0.0	0.0	0.0	*1.1*	*4.4*
	20-29 years	0.2	0.5	2.3	2.8	0.2	0.5	0.8	1.2	0.5	0.9	*4.1*	*5.8*
	30-39 years	1.3	2.0	3.1	2.6	0.7	0.8	1.2	1.2	0.6	0.9	*7.0*	*7.5*
	40-49 years	2.7	2.9	3.4	2.2	1.1	0.9	0.5	0.4	1.4	1.5	*9.0*	*7.9*
	50-59 years	2.0	2.0	3.6	2.2	2.9	2.1	2.2	1.4	1.2	1.1	*11.9*	*8.8*
	60-69 years	1.3	1.3	4.3	2.4	3.6	2.7	0.6	0.4	0.5	0.5	*10.4*	*7.3*
	70+ years	0.6	1.0	2.2	2.0	1.6	1.9	2.4	2.4	1.4	2.2	*8.2*	*9.4*
Male	**< 20 years**	0.2	1.2	0.1	0.3	0.1	0.4	0.0	0.0	0.0	0.0	*0.5*	*1.9*
	20-29 years	1.3	3.0	1.7	2.2	0.2	0.4	1.7	2.5	0.4	0.8	*5.3*	*8.8*
	30-39 years	1.3	2.1	2.9	2.6	0.4	0.4	1.2	1.2	1.3	2.1	*7.1*	*8.4*
	40-49 years	1.4	1.7	3.9	2.6	0.7	0.6	1.4	1.1	1.3	1.6	*8.8*	*7.6*
	50-59 years	1.4	1.6	5.7	3.6	1.3	1.0	2.8	2.0	1.3	1.3	*12.5*	*9.4*
	60-69 years	0.6	0.6	4.9	3.2	1.6	1.4	0.8	0.5	0.2	0.2	*8.2*	*6.0*
	70+ years	0.2	0.4	2.5	2.4	0.7	0.9	1.3	1.4	1.1	1.6	*5.9*	*6.7*
Total		*15.8*	*24.4*	*40.7*	*31.3*	*15.2*	*13.9*	*17.0*	*15.6*	*11.3*	*14.8*	*100.0*	*100.0*

Table 12: Three-way table with shares for gender, age and education comparing raw (unweighted) and adjusted (weighted) data for panel data Wave 2 and Wave 3 (W23)

		Highest completed training (education)											
		Compulsory school		**Apprenticeship**		**Vocational middle school**		**General or vocational secondary school**		**University**		***Total***	
Gender	**Age Groups**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	***raw***	***adjusted***
Female	**< 20 years**	0.4	2.2	0.0	0.0	0.3	0.6	0.6	1.2	0.1	0.3	*1.4*	*4.3*
	20-29 years	0.4	0.9	1.9	1.8	0.4	0.4	1.7	1.8	1.3	2.3	*5.7*	*7.3*
	30-39 years	0.9	1.7	3.7	2.7	1.0	0.9	1.3	1.0	0.7	1.0	*7.5*	*7.3*
	40-49 years	1.9	2.8	3.8	2.1	1.4	1.0	0.7	0.4	1.0	1.0	*8.8*	*7.3*
	50-59 years	1.8	2.6	4.2	2.4	2.8	1.8	1.5	1.0	1.3	1.3	*11.6*	*9.1*
	60-69 years	1.4	2.0	4.8	2.6	2.7	1.7	0.6	0.4	0.4	0.4	*10.0*	*7.0*
	70+ years	0.4	1.1	2.3	2.2	1.5	1.7	2.0	2.1	1.0	1.7	*7.3*	*8.8*
Male	**< 20 years**	0.3	1.5	0.2	0.4	0.0	0.0	0.1	0.2	0.0	0.0	*0.5*	*2.0*
	20-29 years	0.7	1.9	1.3	1.5	0.4	0.6	1.6	1.9	0.4	1.0	*4.6*	*6.9*
	30-39 years	0.9	1.8	4.4	3.7	0.5	0.5	1.0	0.9	1.0	1.6	*7.8*	*8.7*
	40-49 years	1.3	2.1	3.9	2.5	1.2	0.9	1.4	1.0	1.3	1.6	*9.1*	*8.1*
	50-59 years	1.3	2.1	5.8	3.7	1.7	1.3	1.8	1.2	0.7	0.8	*11.3*	*9.1*
	60-69 years	0.7	1.1	5.0	3.1	1.4	1.1	1.2	0.8	0.4	0.5	*8.8*	*6.6*
	70+ years	0.2	0.5	2.4	2.6	1.3	1.7	1.2	1.5	0.5	1.1	*5.7*	*7.4*
Total		*12.6*	*24.5*	*43.8*	*31.3*	*16.7*	*13.9*	*16.7*	*15.5*	*10.2*	*14.7*	*100.0*	*100.0*

Table 13: Three-way table with shares for gender, age and education comparing raw (unweighted) and adjusted (weighted) data for panel data Wave 1 and Wave 2 (W12)

		Highest completed training (education)											
		Compulsory school		**Apprenticeship**		**Vocational middle school**		**General or vocational secondary school**		**University**		***Total***	
Gender	**Age Groups**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	***raw***	***adjusted***
Female	**< 20 years**	0.4	2.7	0.1	0.5	0.0	0.0	0.0	0.0	0.0	0.0	*0.5*	*3.2*
	20-29 years	0.3	0.8	1.6	2.8	0.1	0.4	0.7	1.1	0.5	1.3	*3.2*	*6.3*
	30-39 years	1.5	2.3	2.6	2.3	0.8	0.9	1.2	1.1	0.7	1.0	*6.7*	*7.7*
	40-49 years	2.1	2.5	3.6	2.5	1.2	1.0	0.5	0.4	1.5	1.5	*9.0*	*7.9*
	50-59 years	2.3	2.2	3.4	2.1	3.1	2.1	2.4	1.5	1.3	1.2	*12.5*	*9.1*
	60-69 years	1.5	1.4	4.8	2.5	3.9	2.6	0.7	0.4	0.5	0.6	*11.4*	*7.5*
	70+ years	0.7	1.1	2.3	2.2	1.5	1.7	2.4	2.4	1.3	1.9	*8.2*	*9.4*
Male	**< 20 years**	0.3	2.1	0.0	0.0	0.1	0.7	0.0	0.0	0.0	0.0	*0.4*	*2.8*
	20-29 years	0.8	2.3	1.3	2.3	0.1	0.3	1.7	3.2	0.1	0.4	*4.2*	*8.3*
	30-39 years	1.5	2.3	3.1	2.8	0.4	0.4	0.8	0.7	1.3	2.1	*7.1*	*8.3*
	40-49 years	1.5	1.7	3.5	2.4	0.8	0.7	1.6	1.2	1.5	1.7	*8.9*	*7.6*
	50-59 years	1.6	1.7	5.8	3.5	1.3	1.0	2.7	1.7	1.5	1.4	*12.9*	*9.2*
	60-69 years	0.7	0.6	5.1	3.1	1.7	1.3	0.9	0.5	0.3	0.2	*8.7*	*5.8*
	70+ years	0.3	0.4	2.8	2.6	0.8	0.9	1.3	1.3	1.1	1.5	*6.3*	*6.8*
Total		*15.3*	*24.1*	*40.0*	*31.4*	*16.0*	*14.0*	*17.0*	*15.7*	*11.7*	*14.8*	*100.0*	*100.0*

4. Additional variables prepared for the dataset users

4.1. The International Standard Classification of Occupations (ISCO-08) in VIC III

The third wave also includes the ISCO-08 classification for the respondents and – if available – also for their partners. ISCO-08 is an updated version of the ISCO-88 classification. This update was necessary because the ISCO-88 classification was not able to accurately represent the job market in the face of the rapid development in the information and communication technology sector. Also, the ISCO-08 provides a more detailed picture of the occupational structure in a given country. A comprehensive list for the categories and their definitions can be found at the website of the International Labor Organization.[14] [15]

In the third wave of the VIC survey, respondents were asked to state the name of their current or past occupation as open-end questions (VICAT3_Q18 and VICAT3_Q19A2). Additionally, the respondents shared their formal educational level (VIC3_Q13). The same information was inquired about the respondents' partners (VICAT3_Q28, VICAT3_RQ29 and VICAT3_Q16). Using this information, we categorized each of the respondents' open-end answers into the ISCO-08 categories by hand.[16] The basis for assigning respondents to their respective groups was a document from Statistics Austria that gives detailed accounts about the typed of occupations for each (sub)group for the case of Austria.[17]

Missing values are coded with 9999, answers that could not be assigned to any group are coded with 9998. Respondents that are still in training or education are coded as -9999. In addition to that, the variables in VIC III also feature the category -9998. These are used for respondents that could be categorized in the previous wave, but who could not be classified based on their answers in VIC III. Users are free to carry over respondents' categories from VIC II into VIC III if they like, but we provide the pure data without such imputations.

The ISCO-08 classification comes in four different types. The broadest typification (single digit) divides the respondents into ten major groups (see Table 14). Each additional digit further differentiates them. The second digit adds information about the occupation group (e.g., in group 5: personal service workers vs. sales workers), the third digit gives information about which occupational subgroup the respondent belongs to (e.g., in group 51: cooks vs. waiters and bartenders). The ISCO-08 codes are often used to calculate indices regarding socioeconomic positions. For this, we recommend

[14] ISCO - International Standard Classification of Occupations (ilo.org) (last accessed 08/18/2021)
[15] Bureau of Statistics, work unit of the Policy Integration Department (ilo.org) (last accessed 08/18/2021)
[16] Züll, C. (2015). Berufscodierung. Mannheim, GESIS Leibniz Institut für Sozialwissenschaften (GESIS Survey Guidelines). https://doi.org/10.15465/gesis-sg_019
[17] Vgl. Statistik Austria Erläuterungen zu ISCED 08 (last accessed 08/18/2021)

using the SPSS-syntax files by Ganzeboom (2010).[18] [19] The third wave of the VIC panel provides the one-, two-, and three-digit versions of the ISCO-08 classification for the respondents and their partners. The higher the number of digits in the ISCO-08 classification, the more cases are coded as 9998. The reason for that is that while the information a respondent gives is sufficient to classify them in the single-digit variable, their answer may not be descriptive or unambiguous enough to do the same in the two-, or three-digit variable.

Also, the ISCO-08 codes in VIC III differ from the codes in VIC II in some cases. There are two reasons for this. First, respondents could have changed their profession which would naturally result in different ISCO codes. Second, respondents often gave more detailed descriptions of their profession in VIC III than in VIC II. This resulted in some cases that had to be categorized differently in VIC III than in the previous wave due to the new information that these respondents provided.

Table 14: Ten major groups of the ISCO-08 classification (single-digit)

ISCO-08 (single-digit)	Types of occupations in this group
0	Armed forces occupations
1	Managers
2	Professionals
3	Technicians and associate professionals
4	Clerical support workers
5	Service and sales workers
6	Skilled agricultural, forestry and fishery workers
7	Craft and related trades workers
8	Plant and machine operators, and assemblers
9	Elementary occupations

4.2. The Inglehart Index

The VIC study is devoted to the study of values in times of crisis. Thus, it is of major relevance to conceptualize well-functioning measurements of values which enable sophisticated conclusions of value dynamics in the midst of the crisis. One of the first measurements which was integrated in the World Value Survey from the beginning, was the four-item scale of value priorities by Inglehart (1979)[20], it is called the Inglehart Index.

In the form of a ranking procedure, two materialistic and two post-materialistic goals are given, and the respondents are asked to name one goal of the first and one of second priority (Inglehart & Abramson, 1999)[21]. Based on the judgements of the respondents it is possible to assign individuals to

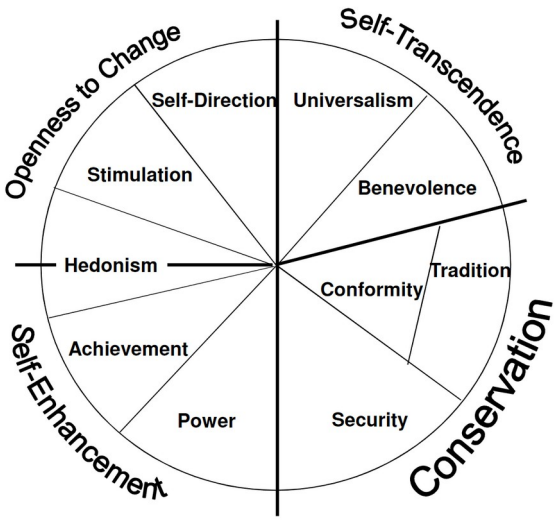
[18] HARRY GANZEBOOM'S Tools for occupation status measures for ISCO-88 (last accessed 08/18/2021)
[19] Ganzeboom, H. B. G. (2010). A new international socio-economic index (ISEI) of occupational status for the international standard classification of occupation 2008 (ISCO-08) constructed with data from 11 the ISSP 2002-2007. Paper presented at the Annual Conference of International Social Survey Programme, Lisbon.
[20] Inglehart, R. (1979). *Value priorities and socioeconomic change.* In S. Barnes & M. Kaase (Eds.), Political action: Mass participation in five Western democracies (S. 305–342). Beverly Hills: Sage Publications.
[21] Inglehart, R., & Abramson, P. R. (1999). Measuring postmaterialism. *American Political Science Review*, 93 (3), 665-677. https://doi.org/10.2307/2585581

materialists, post-materialists and mixed types. A standard variable is included in all three waves which allows comparing the proportion of people being clearly materialist, favouring materialism, favouring post-materialism and falling into the category of clear postmaterialists. Please note that the measurement might be biased in wave 3 because one of the four Inglehart items is "fight against rising prices". In times of high inflation rates (the fieldwork period of July 2022 already reflected the current inflation crisis) this measurement might be biased and might lead to a disproportionally high share of materialists.

4.3. The Schwartz values

In addition, VIC includes the value concept of Shalom Schwartz (1992)[22] measuring 10 individual value orientations. Schwartz has developed extensive scales trying to measure explicitly people's basic values. The ten individual value orientations form a continuum with incompatible values opposing each other in the circular structure and values highlighting similar goals being adjacent. In addition, Schwartz formed two superordinate bipolar dimensions. In the dimension of openness to change vs. conservation (simply put: individualism vs. conservatism), values that push individual thinking and action or emphasize the preservation of security and tradition are opposed to each other. The second bipolar dimension of self-enhancement vs. self-transcendence (simply expressed: egoism vs. altruism) describes the striving for one's own success and dominance vs. the acceptance of others as equal.

Figure 2: The classical circumplex structure of values according to Schwartz (Schwartz, 2012) [23]



[22] Schwartz, S. H. (1992). Universals in the Content and Structure of Values: Theoretical Advances and Empirical Tests in 20 Countries. *Advances in Experimental Social Psychology*, 25, 1–65. https://doi.org/10.1016/S0065-2601(08)60281-6

[23] Schwartz, S. H. (2012). An Overview of the Schwartz Theory of Basic Values. *Online Readings in Psychology and Culture*, 2(1). https://doi.org/10.9707/2307-0919.1116

After establishing his value theory Schwartz et al. (2001)[24] developed a new scale, which is more suitable for representative population surveys, called the Portrait Value Questionnaire (PVQ). This scale consists of 40 short verbal portraits of individuals. The portraits indicate the goals and expectations of the individuals and thus implicitly address the importance of their values. The respondent must indicate on a six-point scale how similar he or she feels to the person.

For the VIC study, the shortened version of the PVQ (with 22 indicators) was used, which was also implemented in the European Social Survey, one of the major survey instruments in Europe. Although this scale is also rather limited, because each value is just measured by two to three items, the scale provides a comprehensive measurement of individual value orientations. It often makes sense to calculate higher-order values and to integrate the higher-order dimensions to come to more reliable conclusions. That is the reason why the data contains different potential value measurements out of the Schwartz-scale. The ten value orientations were calculated adopting the scale use correction because it is important to correct for individual and cultural differences in scale use. We fully followed the procedures to come to reliable scales which are recommended by Schwartz, Breyer & Danner (2015)[25]. The indices of the ten individual values were used to create the higher order values and two bipolar dimensions of openness to change vs. conservation of order and self-enhancement vs. self-transcendence. The two opposite dimensions (which usually form a sufficient negative correlation) were subtracted from each other. All those value scores are available for all three waves of the survey.

4.4. The measurement of personality traits

Besides common value instruments, the VIC 1 questionnaire captures personality scales. Table 10 gives an overview of the measurement of the key personality traits in the survey.

Table 15: The 10-item based Big Five Inventory (BFI-10) and the empathy measurement used in the source questionnaire

[24] Schwartz, S.H., et al. (2001). Extending the cross-cultural validity of the theory of basic human values with a different method of measurement. *Journal of Cross-Cultural Psychology*, 32, 519–542. https://doi.org/10.1177/0022022101032005001

[25] Schwartz, S. H., Breyer, B., & Danner, D. (2015). Human Values Scale (ESS). Zusammenstellung sozialwissenschaftlicher Items und Skalen (ZIS). https://doi.org/10.6102/zis234 (last accessed 01/09/2024)

Personality Traits	Items	Pole	Based on
Extraversion	I am someone who is reserved.	-	BFI 10 (Rammstedt & John, 2007), two items per dimension with five-point scale from 1 = disagree strongly to 5 = agree strongly)
	I am someone who is outgoing, sociable.	+	
Agreeableness	I am someone who is generally trusting.	+	
	I am someone who tends to find fault with others.	-	
Conscientiousness	I am someone who tends to be lazy.	-	
	I am someone who does a thorough job.	+	
Emotional lability	I am someone who is relaxed, handles stress well.	-	
	I am someone who gets nervous easily.	+	
Openness	I am someone who has few artistic interests.	-	
	I am someone who has an active imagination.	+	
Empathy	I often have tender, concerned feelings for people less fortunate than me.	+	Based on Davis (1983) (selection of 4 out of 28 items, five-point scale from 1 = does not describe me at all to 5 = describes me perfectly well
	When I see people being taken advantage of, I feel kind of protective towards them.	+	
	I am often quite touched by things that I see happen.	+	
	I would describe myself as a pretty soft-hearted person.	+	

One of the well-known concepts in personality research is the BIG-Five approach[26], which states that the human personality consists of five major personality factors. In the survey, the 10-item based Big Five Inventory (BFI-10)[27] represents a short measurement of the five personality traits Extraversion, Agreeableness, Conscientiousness, Emotional Stability, and Openness. The scale was especially designed for multi-thematic questionnaires in which respondents' time is severely limited. Additionally, a few items of the sophisticated measurement of empathy by Davis (1983)[28] were used to calculate an empathy index. This computation is simply the mean of all judgements measuring empathy.

4.5. Political Ideologies (POLID) Scale

The dataset also includes a newly developed scale to measure people's ideological dispositions by Ulrich (2021) named the POLID scale. This scale conceptualized political thought as a trade-off between two economic and two sociocultural ideologies. Hence, the scale includes the four major ideologies of the western world (Economic Libertarianism, Conservatism, Socialism and Liberalism) as their main subdimensions. This means that respondents gather four ideological POLID-Scores that each represent one of the ideologies. To replicate the trade-off between these ideologies, the POLID-Scores are calculated as mean-centered scores.[29]

[26] Costa Jr, P. T., & McCrae, R. R. (1992). The five-factor model of personality and its relevance to personality disorders. *Journal of personality disorders*, 6(4), 343-359. https://doi.org/10.1521/pedi.1992.6.4.343
[27] Rammstedt, B., & John, O. P. (2007). Measuring personality in one minute or less: A 10-item short version of the Big Five Inventory in English and German. *Journal of research in Personality*, 41 (1), 203-212. https://doi.org/10.1016/j.jrp.2006.02.001
[28] Davis, M. H. (1983). Measuring individual differences in empathy: Evidence for a multidimensional approach. *Journal of personality and social psychology*, 44(1), 113-126. https://doi.org/10.1037/0022-3514.44.1.113
[29] Ulrich, M. (2021). Politische Ideologien (POLID). GESIS-Leibniz-Institut Für Sozialwissenschaften. (Zusammenstellung sozialwissenschaftlicher Items und Skalen) https://doi.org/10.6102/zis313

In both waves, the sample was randomly split in half and one of these groups were presented with the scale. Since this randomization was carried out for both waves independently, only 186 respondents filled out the scale in both VIC II and VIC III. However, in wave 2 valid scores are reported for 767 respondents and for 765 respondents in wave 3. Respondents that filled out the scale to quickly to have done so diligently are already set as missing.

In VIC II, all variables to calculate the POLID-Scores are included. However, in VIC III two Socialism items were replaced by others that are not part of Ulrich's original version of the scale. VIC II includes these new items as well, which enables researchers to calculate the same POLID-Scores for VIC II and III (see Table 16). T-Tests show that the resulting POLID-Scores do not differ between the two versions in VIC II. Hence, results will not be biased if one uses the version that deviates from Ulrich (2021).

Users find the POLID-Scores calculated in the original version by Ulrich (2021) in the variables VICAT2_POLIDneo, VICAT2_POLIDkon, VICAT2_POLIDsoz and VICAT2_POLIDlib. The variables for the version that is available for both VIC II and VIC III are named VICAT2_neo, VICAT3_neo, VICAT2_kon, VICAT3_kon, VICAT2_soz, VICAT3_soz, VICAT2_lib and VICAT3_lib.

Table 16: The 16 items of each version of the POLID-Scale (X marks items used in the respective version of the scale)

	Item	Original version by Ulrich (2021)	Version implemented in VIC II and VIC III
Libertarianism (neo)	Justice means that I don't have to pay for people who don't perform as well as I do.	X	X
	There is only an incentive to perform well if differences in income are large enough.	X	X
	I admire people who built up a lot of wealth for themselves the most.	X	X
	If the welfare state provides to well for people, then they have no reason to seek work.	X	X
Conservatism (kon)	The state should focus on preserving our common culture and traditions.	X	X
	It is our common culture that holds this country together.	X	X
	You only lead a fulfilling life if you adhere to our traditions.	X	X
	The most important thing for me is to adhere to our common values.	X	X
Socialism (soz)	Equality means that the welfare state ensures that none is left behind.		X
	It is fair if all groups in society are put in equal positions.	X	
	Equality means that poor and rich, men and women, etc. are put on par at the end of the day.	X	X
	The most important thing one can teach their children is empathy with the ones that have if worse.	X	
	Most income inequalities are unjust because people are equal.	X	X
	If the welfare state does not provide enough for people, businesses could exploit them to easily.		X
Liberalism (lib)	The most important thing for me is that I can lead my life as I see fit – as long as I don't harm anyone else.	X	X
	Society must ensure that everyone can act out their freedom as long as they don't harm anyone else.	X	X
	It would be best for all if everyone could to whatever they liked – as long as they don't harm anyone.	X	X
	Freedom means above all that everyone may life their life as they see fit as long as they don't harm anyone else.	X	X

4.6. Quality checks regarding satisfycing

A rather simple typology of satisficers vs. optimizers was introduced in survey research to build a bridge between survey methodologists and cognitive psychologists.[30] When we think about people who answer surveys it is quite plausible that not all participants are motivated to contribute to science and do their best to answer all sorts of questions (the "optimizers").

[30] Krosnick, J. A. (1991). Response strategies for coping with the cognitive demands of attitude measures in surveys. *Applied cognitive psychology*, *5*(3), 213-236.

A significant proportion of respondents (the satisficers) may take the survey quite easy and try to finish their duty as fast as possible. If we identify those who complete scales very quickly, we can assume a better data quality.

That is the reason why we measured in VIC III the time every survey respondent takes to fill out certain scales. This measure was implemented with the Schwartz items (VIC3_Q70A1 - VIC3_Q70A22), the scale on COVID 19 scepticism (VICAT3_Q57A11 - VICAT3_Q57A11), the scale on group focused enmity VICAT3_Q104A1 - VICAT3_Q104A12) and the Polid ideology scale (VICAT3_Q112A1 - VICAT3_Q112A16). We have calculated reasonable response times in order to be able to assume a serious response to the scale. You find those necessary items to detect the satisficers of the survey also at the end of the data set.[31] If people need less than 60 seconds to answer the Schwartz-scale, less than 30 seconds to answer the COVID 19 items or the scale on group focused enmity and less than 100 seconds to answer the POLID scale they might be deleted from the analysis due to quality reasons.

5. Data quality assessment

To assess data quality, we conducted multiple checks involving age and sex information. Two data sources were utilized for this evaluation: the registration data supplied by online access panel members during their sign-up, and information gathered independently across all three survey waves. In total, we conducted twelve distinct tests. In Figure 1, however, only 11 tests are listed instead of 12. Apparently, the frequency data for sex W1W2 are missing. We compared responses for age and sex separately across different wave pairs (W1W2, W2W3, W1W3) and between individual waves (W1, W2, W3) and the registration data. Any disparities between these comparison points were recorded as anomalies. In the case of sex, discrepancies were evident based on predefined categories. Concerning age, anomalies were identified as any implausible age changes, such as someone becoming younger or aging faster than the time intervals between waves would allow. This process is inherently imprecise for age, as even a minor age increase could potentially be flagged as an anomaly. Consequently, the age test is a more conservative estimate compared to the sex test.

[31] They are called VIC3_Schnellausfüller_Schwartz, VICAT3_Schnellausfüller_Q57, VICAT3_Schnellausfüller_Q104 and VICAT3_Schnellausfüller_Q112.

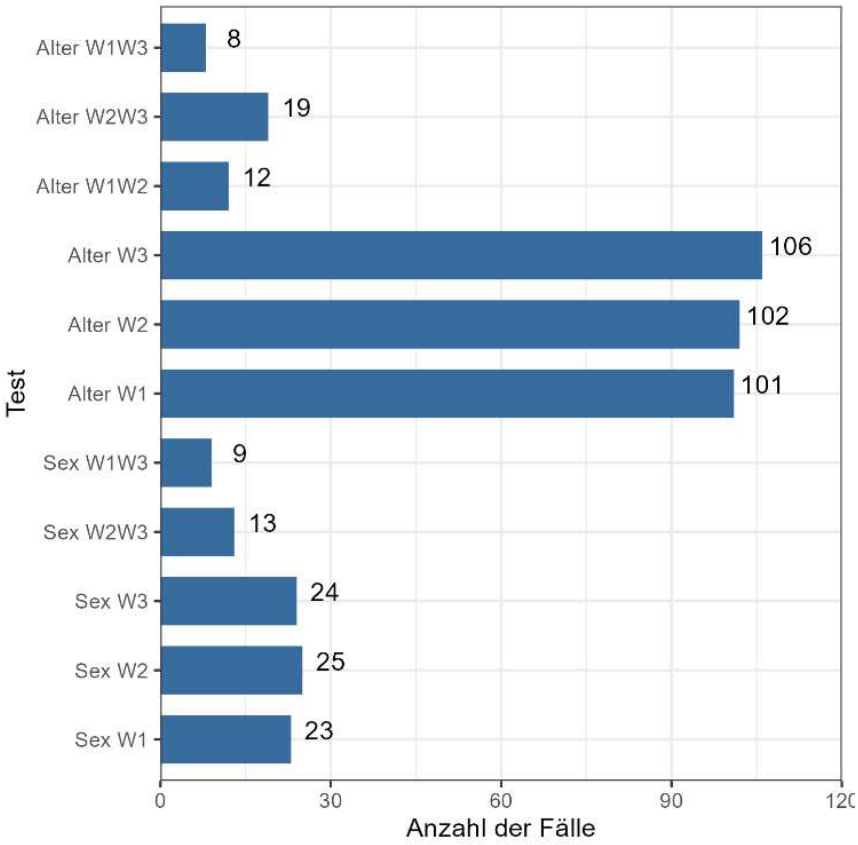


Figure 2: Frequencies of anomalies in the 12 tests

Out of the total 3,797 cases that participated in at least one of the three waves, 3,490 (91.9%) displayed no anomalies. The remaining cases are distributed as shown in Table 17.

Table 17: Sum of anomalies across 12 tests by cases (n=3797)

Sum of anomalies	Cases	Percentage
0	3490	91.91
1	228	6.00
2	46	1.21
3	16	0.42
4	13	0.34
5	2	0.05
6	2	0.05

The dataset includes a variable (Issue_Sum) that provides the sum of anomalies for each record.